Research Progress Presentation 2

PP2

AI-POWERED RESEARCH GAP IDENTIFIER & TREND ANALYZER

R26-SE-023

RESEARCH BACKGROUND

Problem

Millions of papers make manual gap detection impractical, leading to unintentional duplication.

Effect

Waste of research funding and institutional resources due to redundant development.

Solution

Designing a platform that automates the research process using AI driven approach.

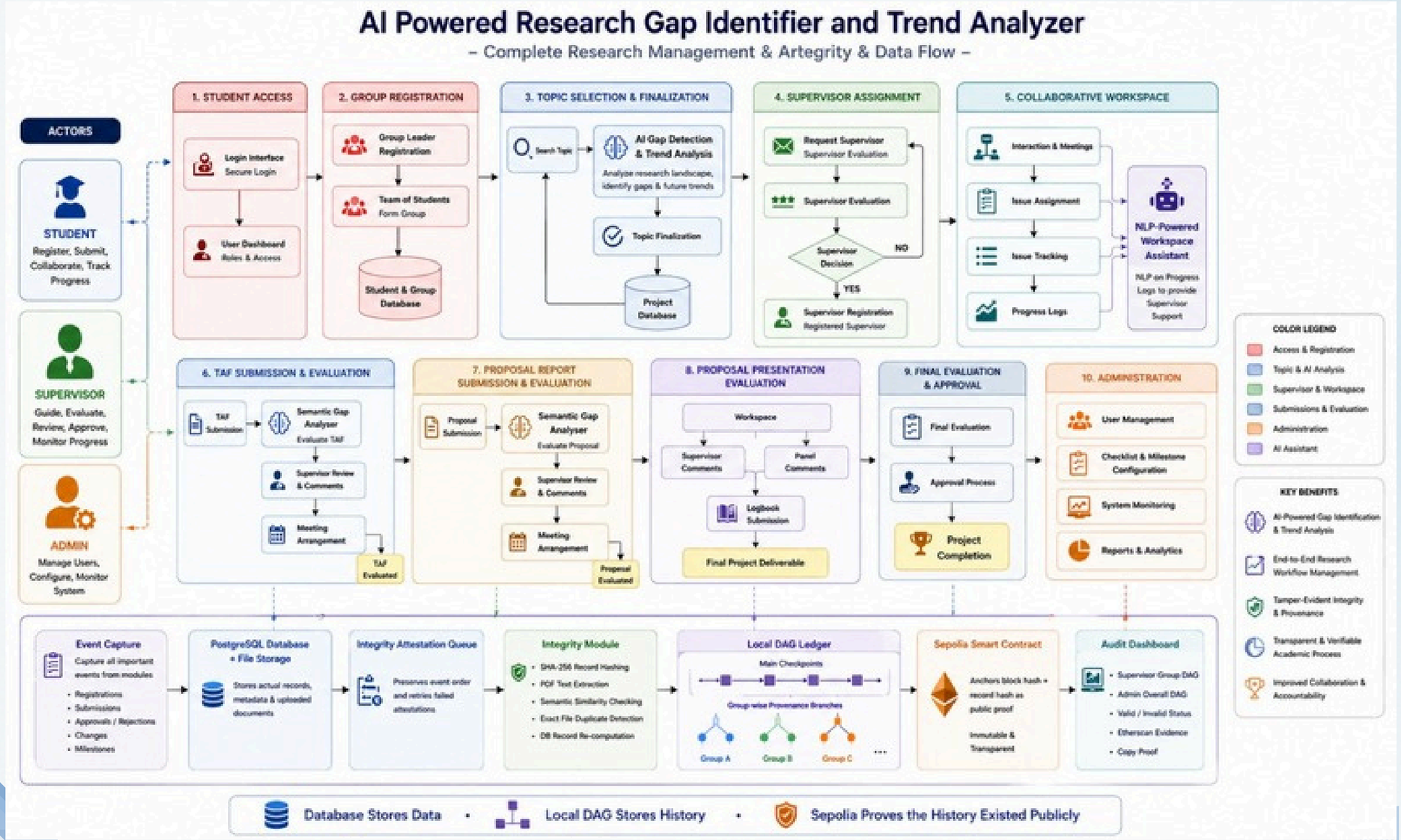
Beneficiaries:

- **Students:** Early validation of novelty.
- **Supervisors:** Quality assurance and automated oversight.
- **Institutions:** Protection of academic reputation and research standards.

COMPONENTS

- Recommendation & Collaboration
- Similarity Analysis & Gap Detection
- Integrity & Governance
- Collaborative Workspace & NPL Powered Supervisor Assistant

FLOW DIAGRAM



RECOMMENDATION AND COLLABORATION



IT22629562

Widanagama K.L.

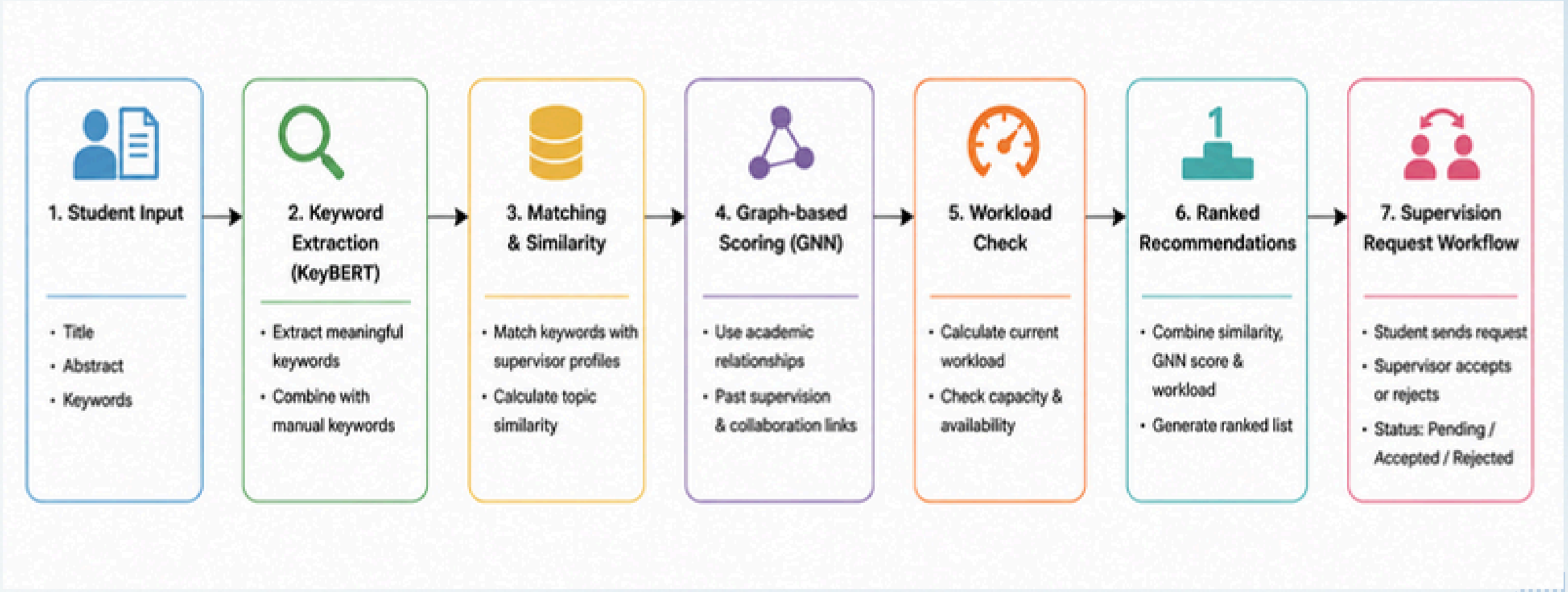
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RECOMMENDATION AND COLLABORATION

- Addresses the **problem** of manual supervisor searching and selection.
- Recommends suitable supervisors based on the research topic submitted by the student.
- Uses **KeyBERT** to extract meaningful keywords from the title, abstract, and keywords.
- Uses a **Graph Neural Network**-based approach to model relationships between supervisors, research topics, past supervision, and collaboration links.
- Considers supervisor workload so that supervisors with high topic match and lower workload are prioritized.

The novelty of the component is using Graph Neural Network to model relationships between supervisors data and topic data.

RECOMMENDATION AND COLLABORATION



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RECOMMENDATION AND COLLABORATION

PP1 Feedback:

- Concern about fully synthetic dataset.
- Synthetic data may not reflect real supervisor expertise accurately.
- Advised to augment an existing dataset instead of creating a complete synthetic dataset.

PP2 Improvement:

- Used publicly available SLIIT academic staff profiles as base data and generated dataset using that.
- Augmented the dataset with research areas, specialties, profile text, collaboration links, and workload fields.
- initialized database with postgreSQL

RESULTS AND DISCUSSION

Matrices

Top-1 Correct	1 if Rank 1 relevance is 1 or 2
Precision@5	Number of relevant supervisors in top 5 / 5
Hit@5	1 if at least one relevant supervisor appears in top 5
NDCG@5 (ormalized Discounted Cumulative Gain)	Are the most relevant supervisors placed near the top of the top 5 recommendation list

RESULTS AND DISCUSSION

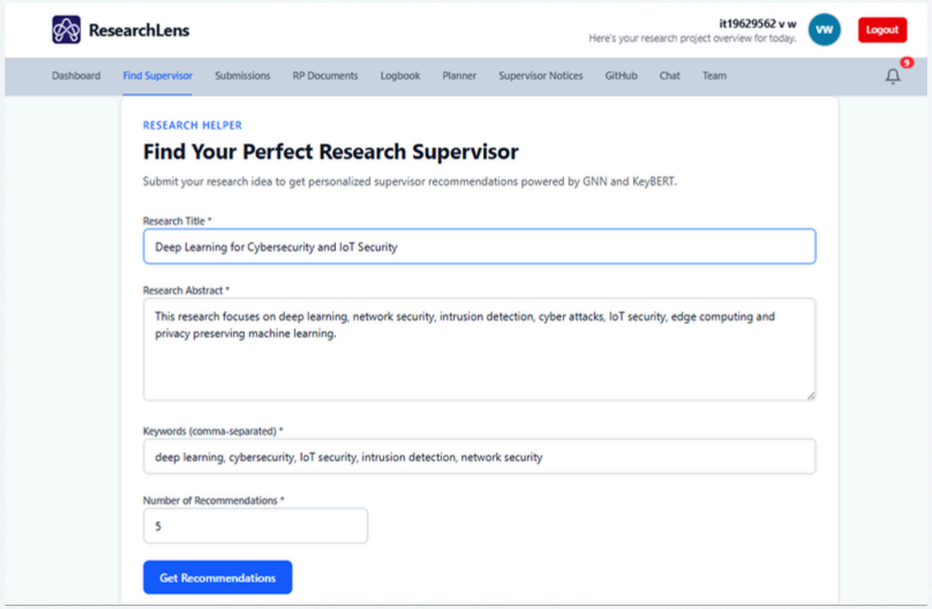
Relevancy

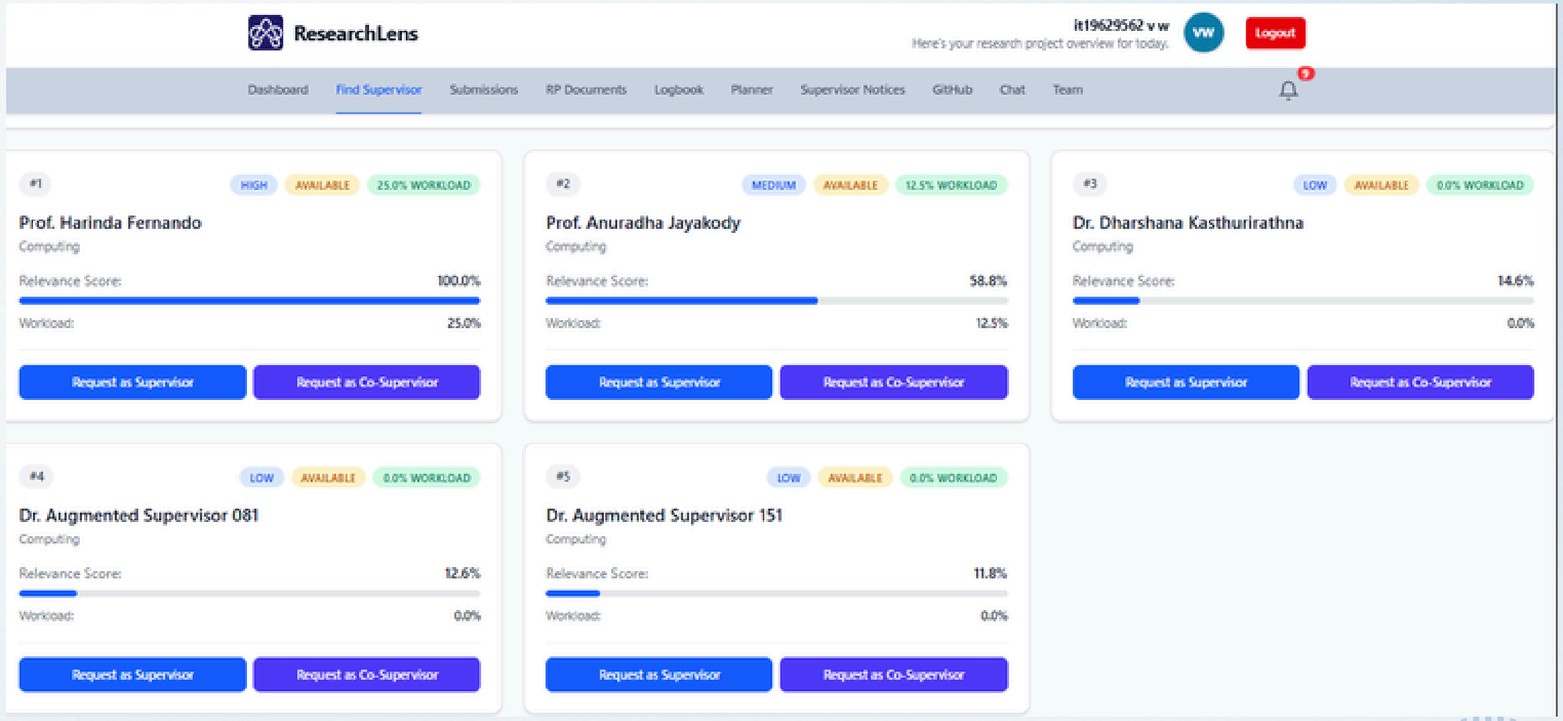
Relevance category	Score	Count
Highly relevant	2	20
Partially relevant	1	69
Not relevant	0	11
Total recommendations	—	100

RESULTS AND DISCUSSION

Metric	Final Result
Number of test topics	20
Top-5 recommendations evaluated	100
Top-1 Relevance Accuracy	100%
Precision@5	89.0%
Hit@5	100%
NDCG@5	94.1%

RESULTS AND DISCUSSION





SIMILARITY ANALYSIS & GAP DETECTION



IT22141842

Bandara S.M.D.M.

B.Sc. (Hons) in Information Technology Specializing in Software Engineering

SIMILARITY ANALYSIS & GAP DETECTION

Problem:

- Validating research ideas against existing work is manual, slow, and subjective.
- No tool tells students if their idea is addresses a genuine gap or overlapping.

Solution:

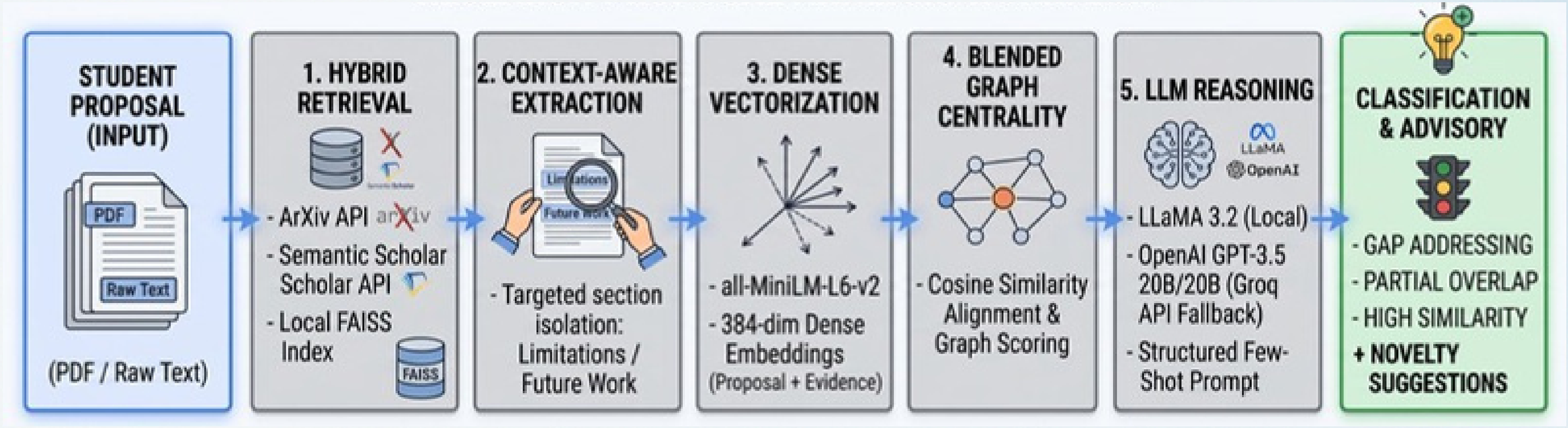
- **5-Stage Pipeline:** Automatically classifies proposals as Gap-Addressing, Partially Overlapping, or Highly Similar to existing work.
- Retrieves live literature from **arXiv** and **Semantic Scholar**, isolates the important sections (Limitations, Future Work) to generate a grounded novelty suggestions.

Novelty:

- **Semantic Meaning** over keyword matching**:** Uses semantic embeddings (all-MiniLM-L6-v2) to detect conceptually similar work even when terminology differs.
- **Grounded LLM:** Output is heavily backed on retrieved evidence, producing explainable, hallucination-free justifications.

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SIMILARITY ANALYSIS & GAP DETECTION



SIMILARITY ANALYSIS & GAP DETECTION

PP1 Feedback:

- **Storing papers** for a specific domain is impractical. Advised to implement a **Crawler** to fetche and cache freely accessible research data dynamically.
- For **paywalled publications**, show potential relevance and note survey limitations so users can self-search.
- Cautioned against **hallucinations** and advised to promote accurate referencing.

PP2 Improvement:

- **Dynamic Global Retrieval:** Live retrieval from arXiv + Semantic Scholar APIs (2.4M+, 200M+ papers) and a lightweight FAISS cache up-to-date literature.
- **Paywalled papers:** Surfaces title, year, and relevance score for paywalled work, and flags full-text as unavailable.
- **Hallucinations:** LLM only reasons over retrieved evidence (grounded RAG), every claim ties back to a shown paper title.

RESULTS AND DISCUSSION

Test Case ID & Domain	Expected vs. Actual	Expected vs. Actual
TC1: Biomedical	Exp: GAP_ADDRESSING Got: GAP_ADDRESSING	69.7%
TC2: Agricultural AI	Exp: HIGH_SIMILARITY Got: HIGH_SIMILARITY	81.2%
TC3: Legal NLP	Exp: PARTIAL_OVERLAP Got: GAP_ADDRESSING	76.1%
TC4: Low-Resource Speech	Exp: PARTIAL_OVERLAP Got: PARTIAL_OVERLAP	62.5%
TC5: Clinical EHR (Lexical Shift)	Exp: GAP_ADDRESSING Got: GAP_ADDRESSING	77.4%

RESULTS AND DISCUSSION

Pipeline Variant / Configuration	Expected vs. Actual	Strict Accuracy ($N=12$)
Baseline 1: Pure TF-IDF	Deterministic (Local)	25.0% (3/12)
Variant C: w/o LLM (Heuristic Only)	Cosine Thresholds	33.3% (4/12)
Variant A: w/o Context Extraction (Naive RAG)	Raw 500-word Chunks	58.3% (7/12)
Variant B: w/o Blended Centrality (Pure Sim)	Ollama (LLaMA 3.2)	50.0% – 58.3%
Proposed System (Groq Cloud)	OpenAI GPT-OSS 20B	41.7% (5/12)
Proposed Full System (Local Ollama)	LLaMA 3.2 (Local)	58.3% (7/12)

INTEGRITY & GOVERNANCE



IT22097156

Darshan R

B.Sc. (Hons) in Information Technology Specializing in Software Engineering

INTEGRITY & GOVERNANCE — RESEARCH PROVENANCE, OWNERSHIP & ACCOUNTABILITY

Problem

- Academic records and decisions can be modified or deleted after submission
- Local database logs alone cannot provide independent proof of history
- Submission reuse/similarity creates additional integrity concerns

Proposed Approach

- Hybrid Tamper-Evident Governance Framework
- Semantic Similarity + Cryptographic Hashing + Local DAG Ledger + Ethereum Sepolia Anchoring

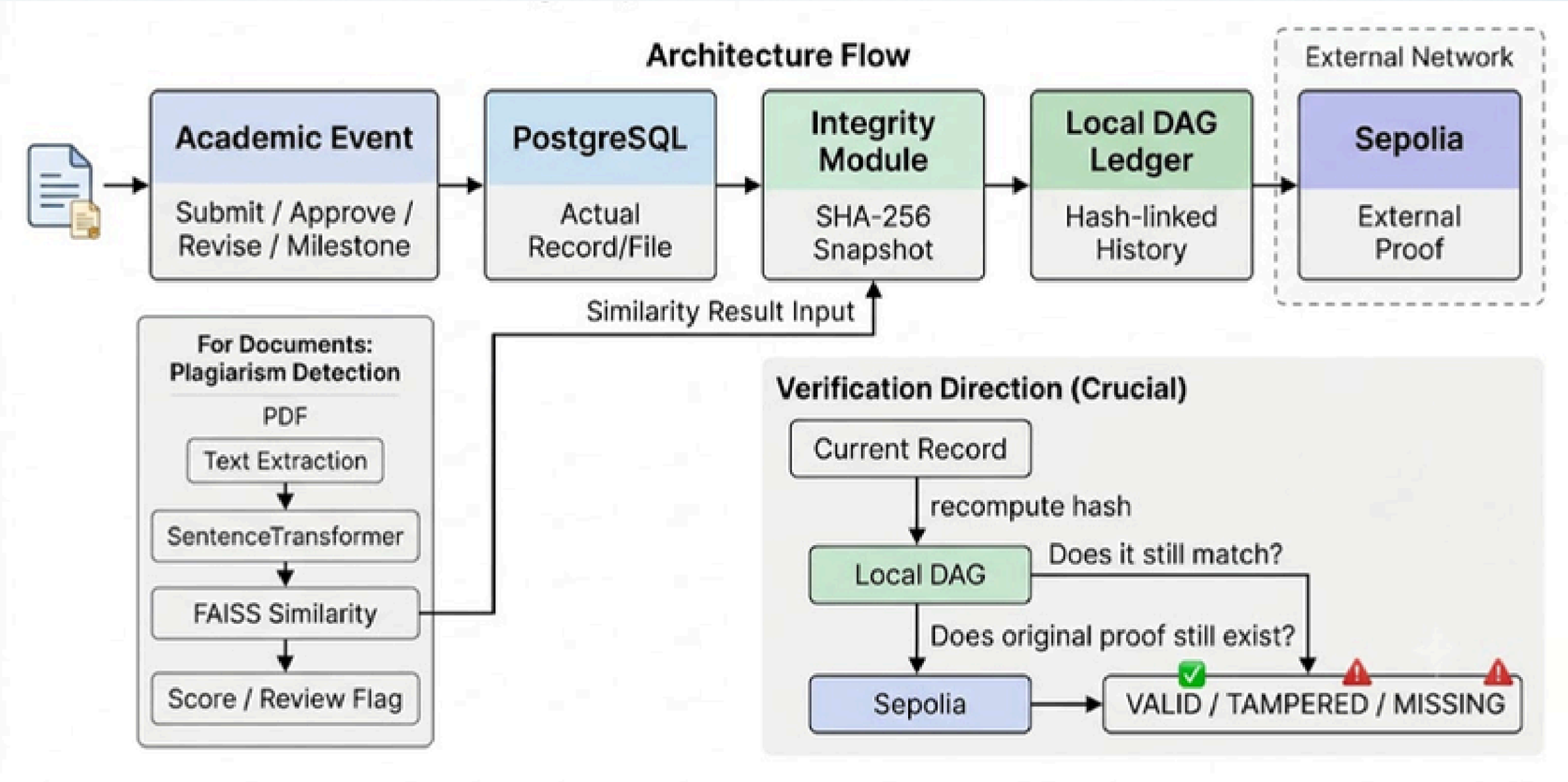
Provides

- Provenance — what happened and when
- Integrity — Detect modification or deletion
- Accountability — Preserve verifiable decisions and contributions
- External Proof — Verify history independently of the local database

The database stores the data, the DAG preserves the history, and Sepolia proves that history existed.

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INTEGRITY & GOVERNANCE — RESEARCH PROVENANCE, OWNERSHIP & ACCOUNTABILITY



Local attestation provides immediate integrity; Sepolia provides independent, externally verifiable evidence.

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INTEGRITY & GOVERNANCE — RESEARCH PROVENANCE, OWNERSHIP & ACCOUNTABILITY

PP1 Feedback

- “Is checking similarity of the submitted TAF and proposal enough? What is the approach? What is the novelty?”

PP1 Interpretation

- Similarity alone is insufficient
- Integrity approach needed clarification
- Need evidence of post-submission changes
- Need independent evidence
- Novelty needed clarification

PP2 Refinement

- Similarity similarity using SentenceTransformer & FAISS retained as a supporting validation layer alongside hash-based duplicate detection.
- SHA-256 → Local DAG Provenance → Sepolia Anchoring provides tamper-evident and independently verifiable history.
- Added recomputation and verification against preserved ledger proofs
- Added Sepolia smart-contract anchoring outside the local database trust boundary
- Extended protection to submissions, approvals, changes & milestones, with independent tamper verification rather than only document comparison.

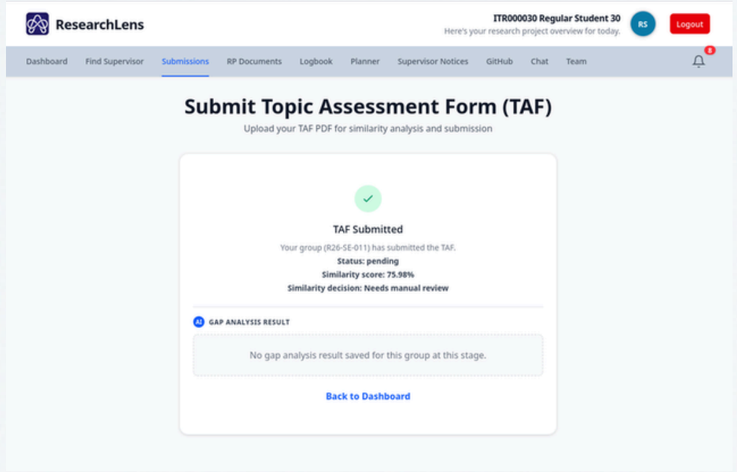
Novelty

Hybrid academic integrity architecture combining semantic validation, tamper-evident DAG provenance, and external blockchain anchoring.

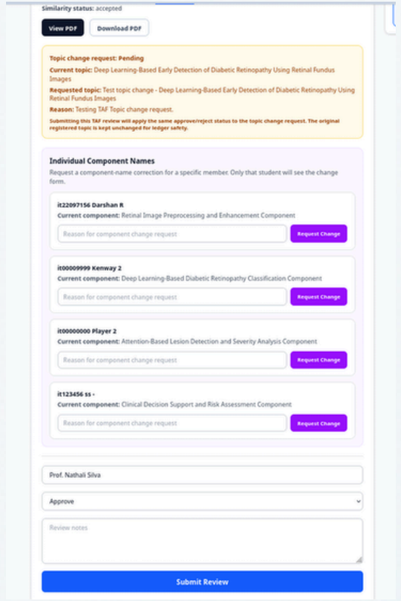
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INTEGRITY & GOVERNANCE — RESEARCH PROVENANCE, OWNERSHIP & ACCOUNTABILITY

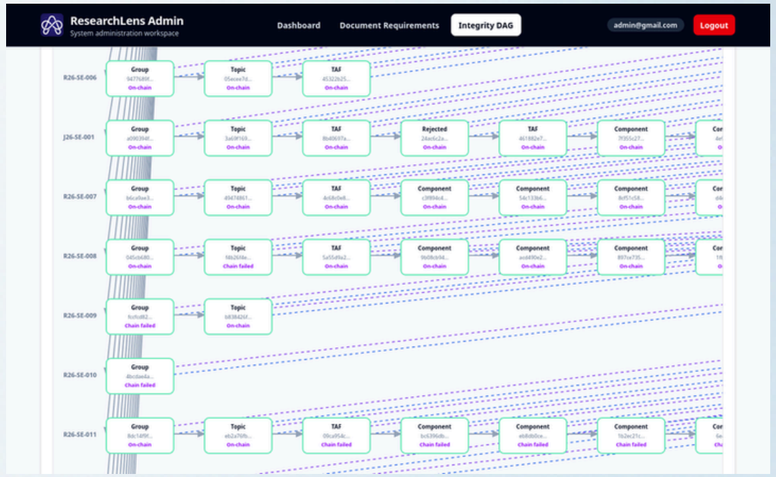
01 - Submission + Similarity



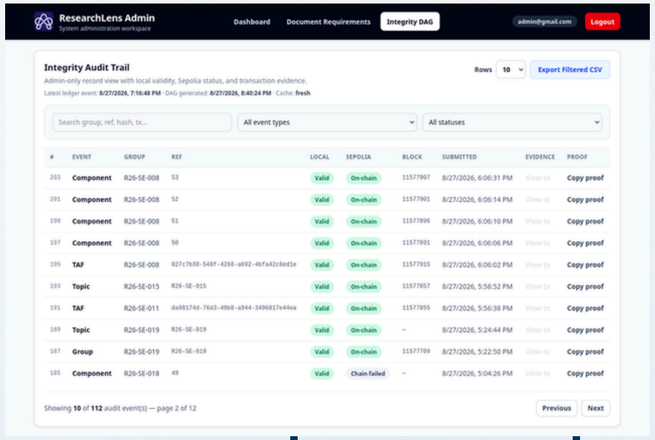
02 — Supervisor Approval / Governance

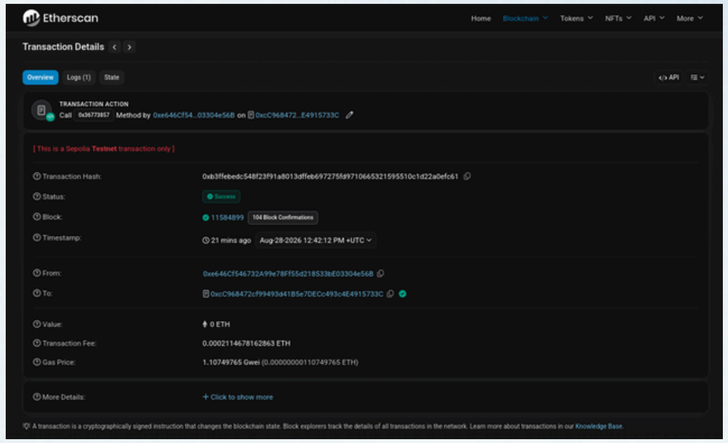


03 - Integrity DAG



04 - Blockchain Evidence





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COLLABORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT



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COLLOBORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT

The Problem

- Academic research supervisors overseeing multiple student groups face a significant review burden manually shifting through lengthy, narrative-heavy research logbooks
- Existing learning analytics and tracking dashboards analyze structured quantitative traces (activity logs, task states) but cannot synthesize unstructured textual progress, obstacles, and methodological reasoning.

Research Gap

- Lack of integrated collaborative supervision platforms that directly combine structured workflow coordination with Transformer-based NLP abstractive summarization while maintaining human-in-the-loop oversight.

Novelty

- An AI-assisted collaborative platform integrating a fine-tuned BART (Bidirectional and Auto-Regressive Transformers) model directly into the student-supervisor workflow
- Human-AI Decision Support: Automated summaries provide an instant high-level overview on the supervisor dashboard, while retaining full access to original logbooks for detailed verification and feedback

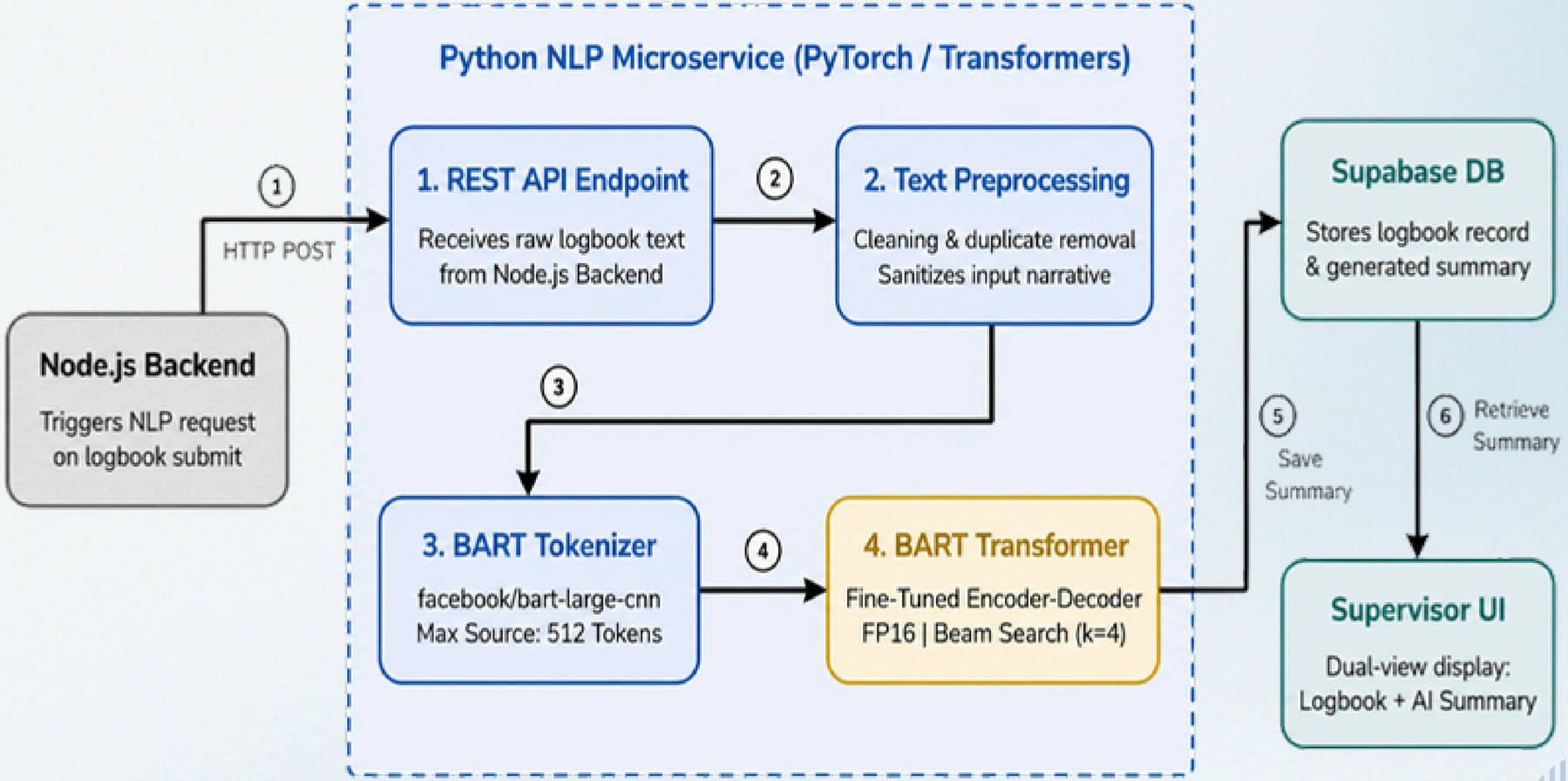
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COLLOBORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT

NPL MICROSERVICE ARCHITECTURE & TECHNICAL IMPLEMENTATION

NLP Microservice

- Python service executing the fine-tuned BART Transformer model, decoupled for independent scaling and evaluation.



NLP Microservice Pipeline Architecture (BART Summarization)

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COLLOBORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT

PP1 Feedback

- Used synthetic data
- Still have the concern regarding the approach and what exactly doing

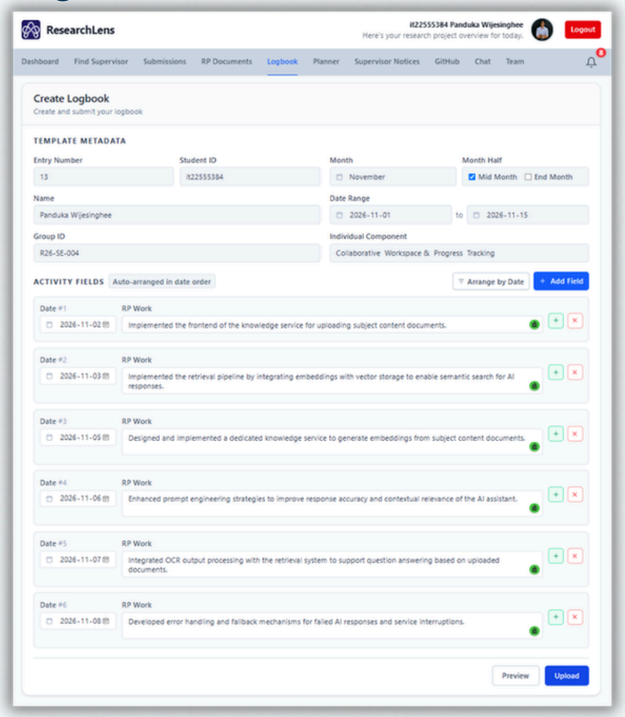
PP2 Improvements

- **Fine-tuned** BART for automated research **Logbook summarization**.
- Evaluated the **fine-tuned** model using ROUGE-1, ROUGE-2 and ROUGE-L metrics.
- Developed **Student Dashboard** submission overview.
- Developed **Supervisor Dashboard** with categorized approved research groups.
- Integrated Microsoft Planner for research **Task management** and **Gruop work management** section.
- Integrated **GitHub Analysis** for development activity monitoring.
- Implemented real-time **Student-Supervisor Chat**.
- Enhanced **Supervisor Notices** section.
- Created **Team Profile** and **User Profile** sections.
- Implemented a dedicated **Notifications** section.
- Added automatic supervisor **Feedback Sggestions** based on generated student logbook summaries.

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RESULTS AND DISCUSSION

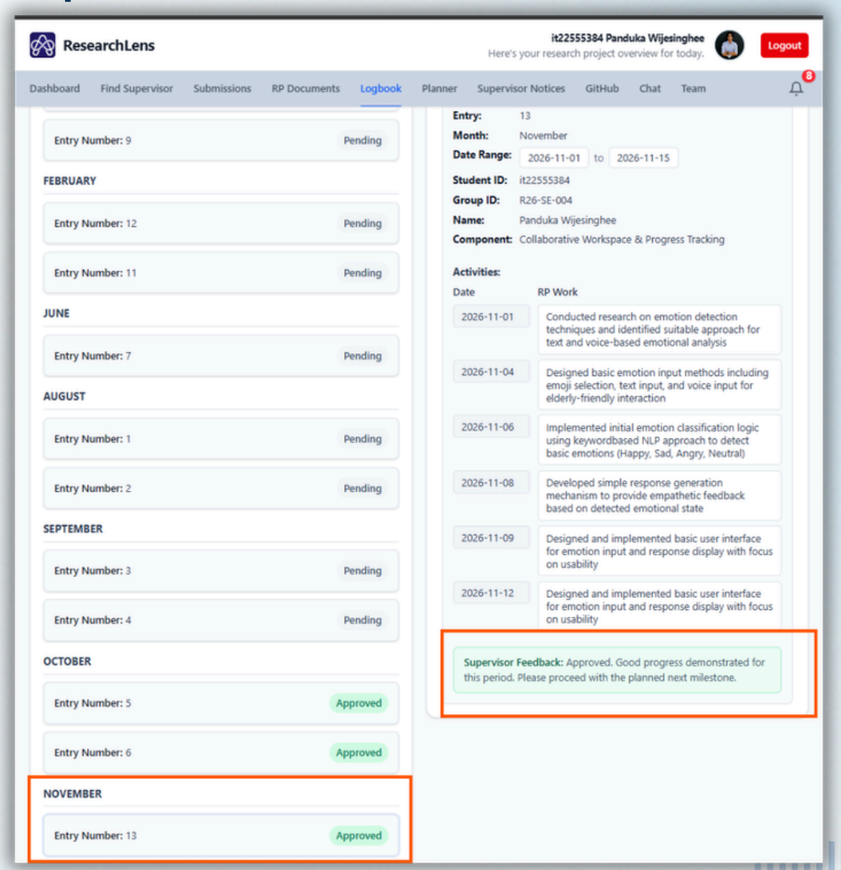
Logbooks Submission



Supervisor reviewing & logbook summary generation

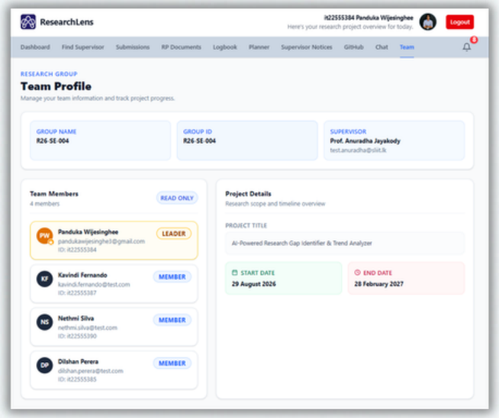


Supervisor Feedback

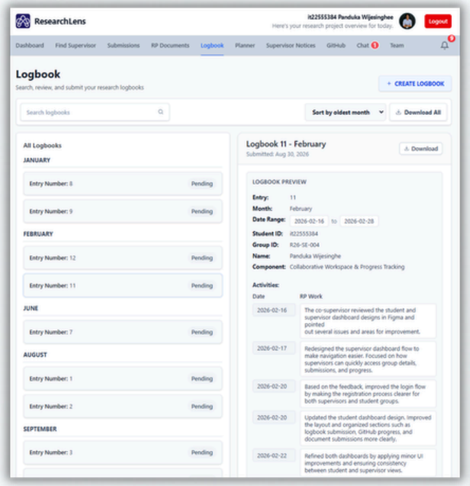


COLLOBORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT

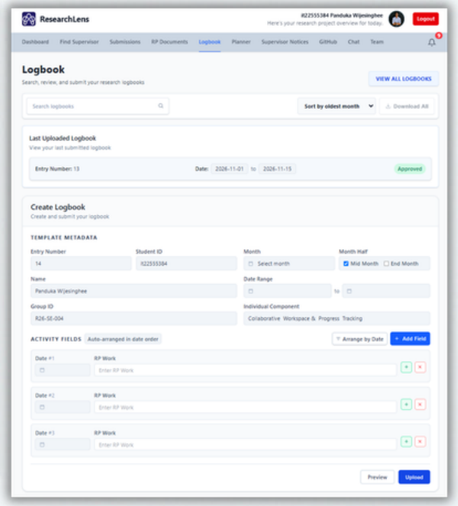
Students Profile



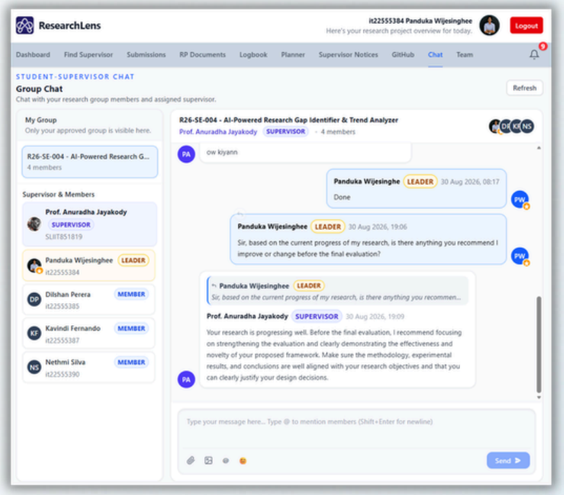
Logbooks section



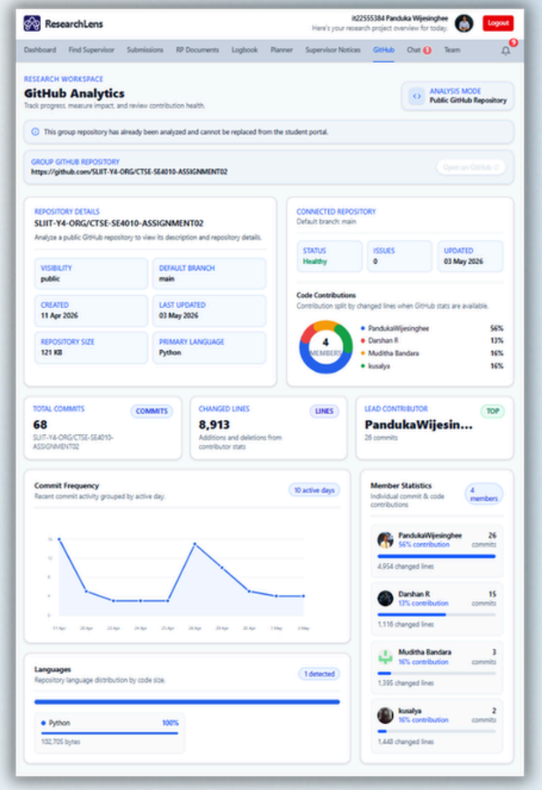
Logbooks Submission



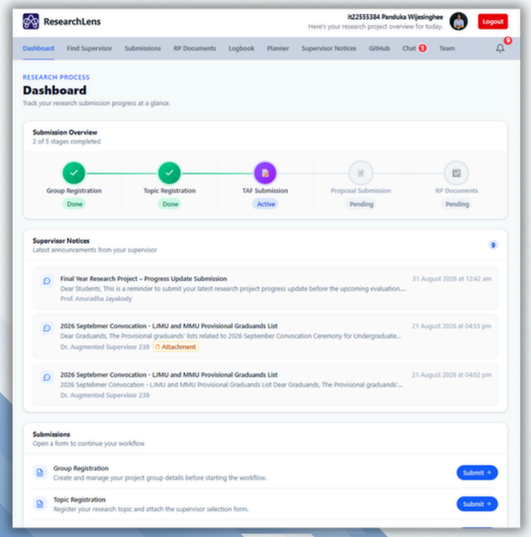
Chat section



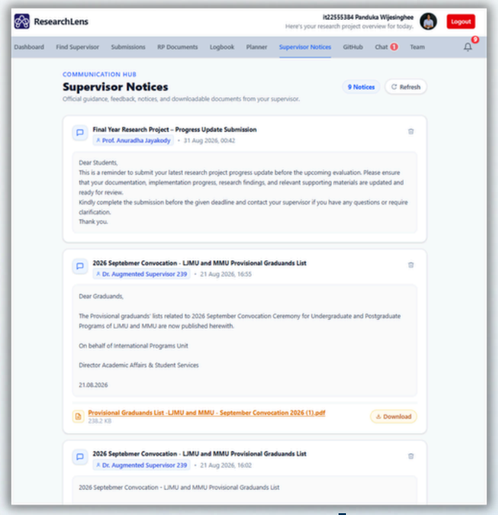
GitHub Analytics



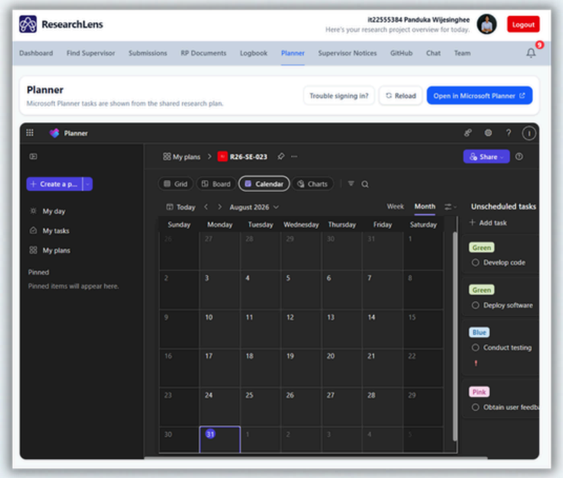
Student Dashboard



Supervisor Announcements



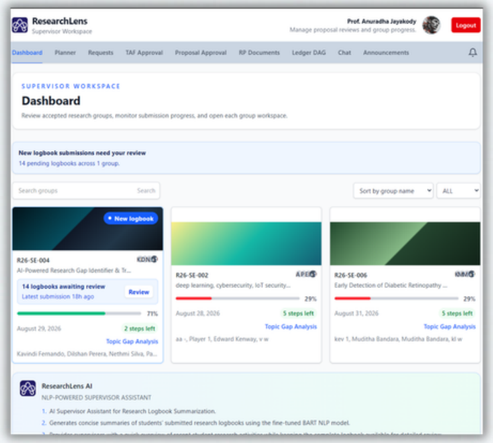
Planner



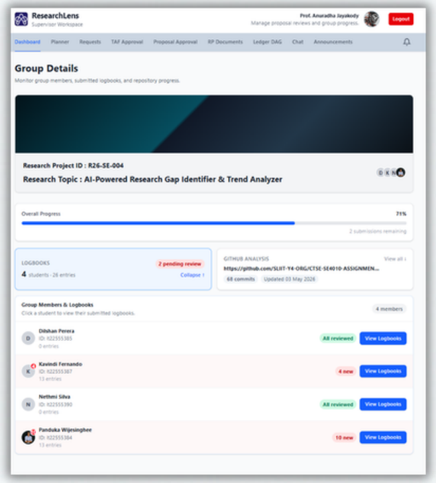
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COLLOBORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT

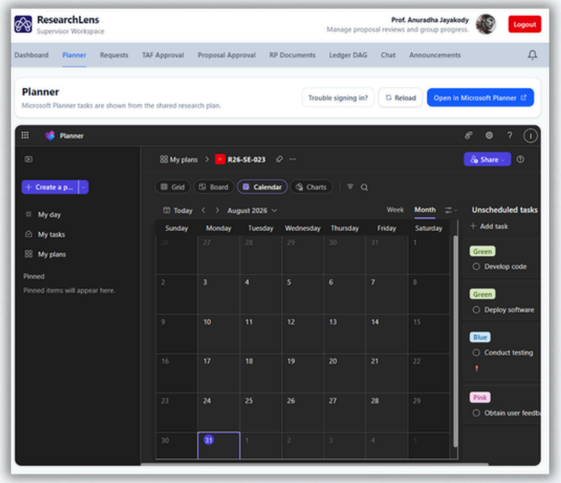
Supervisor Dashboard



Group's Dashboard



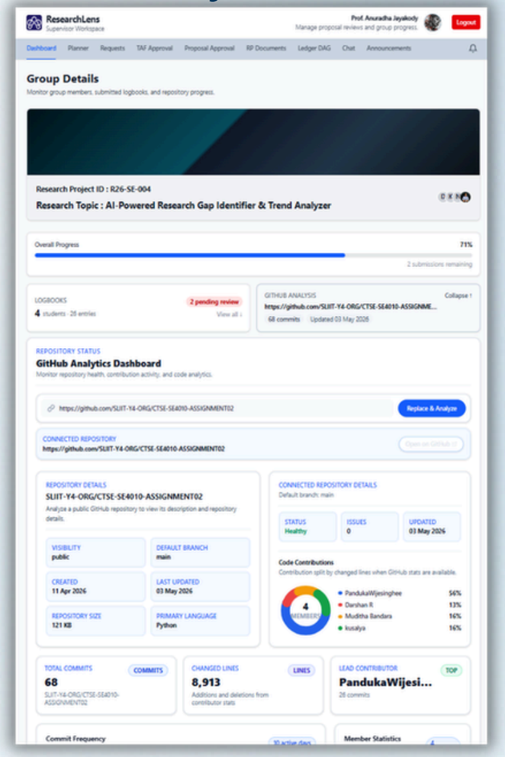
Planner



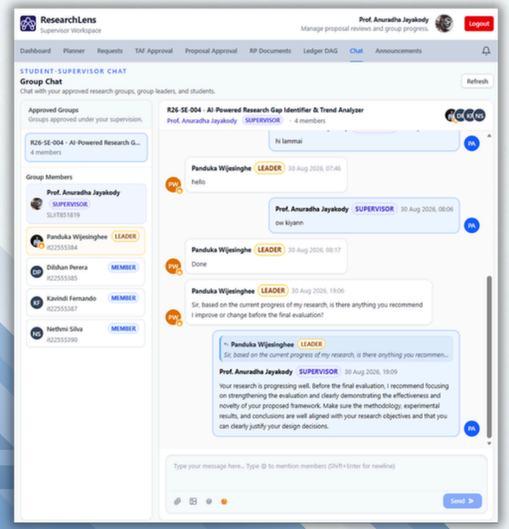
Logbook review section



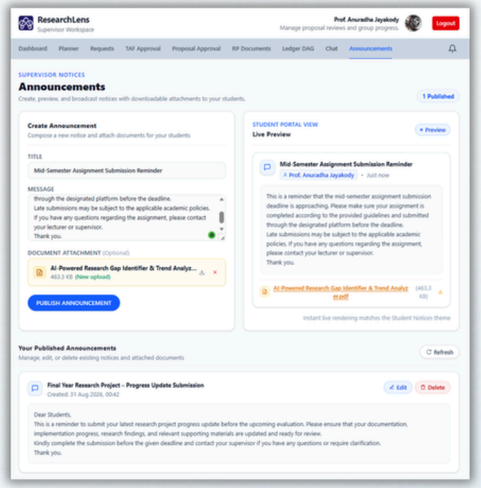
GitHub Analytics



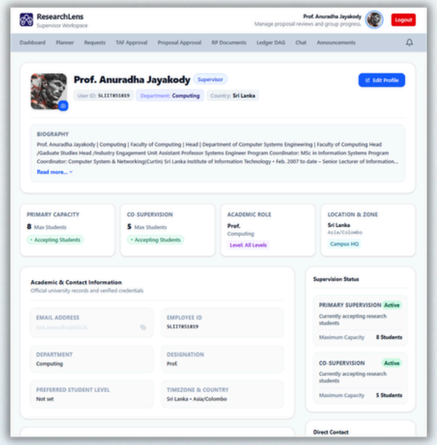
Supervisor Chat section



Supervisor Announcements



Supervisor Profile



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BUSINESS POTENTIAL, COMMERCIALIZATION & ETHICAL CONSIDERATIONS

- **Commercialization & Market Potential:**

Target Audience: Higher Education Institutions (HEIs), university research departments, and postgraduate supervision programs.

Value Proposition: Streamlines communication between students and supervisors while reducing manual review time by providing decision-support summaries.

Business Model: Potential B2B/SaaS platform licensed to universities per department or active student research group.

- **Ethical, Legal & Social Dimensions:**

Data Privacy: Role-Based Access Control (RBAC) via Supabase secures group records, logbooks, and feedback between students and supervisors.

Academic Oversight: AI acts purely as an assistive decision-support tool; academic evaluation, judgment, and feedback remain 100% human-driven.

THANK YOU

GINYARD INTERNATIONAL

I appreciate your interest and hope this presentation has provided valuable insights.

RESULTS AND DISCUSSION

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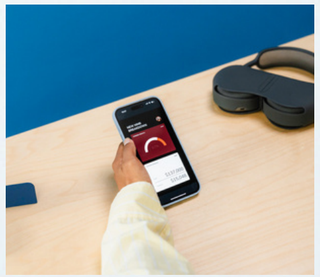
First Results

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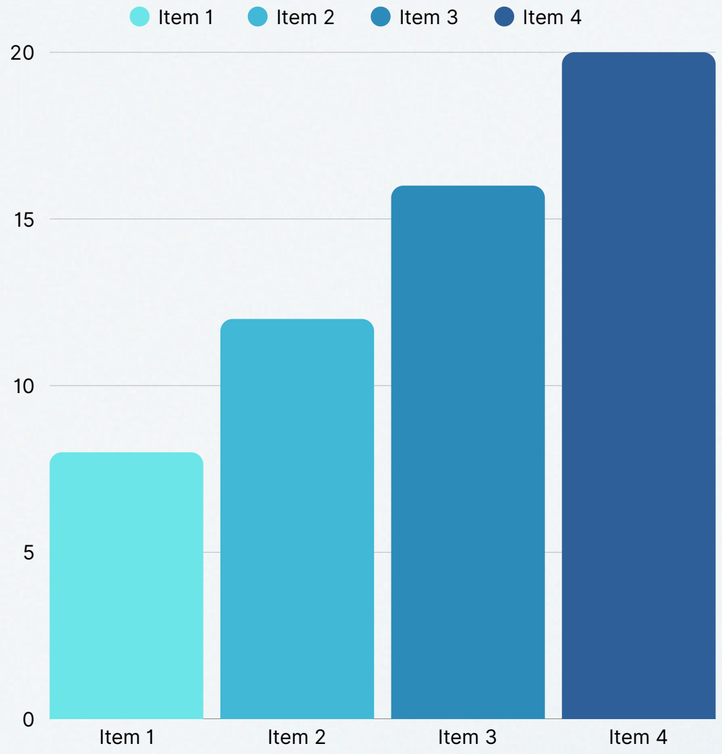
Second Research

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Third Research

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DATA COLLECTION AND ANALYSIS

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RESULTS AND DISCUSSION

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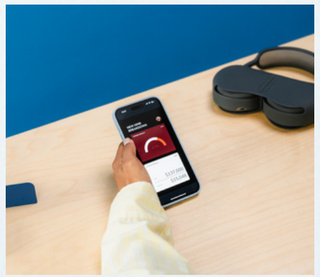
First Results

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Second Research

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Third Research

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RESULTS AND DISCUSSION

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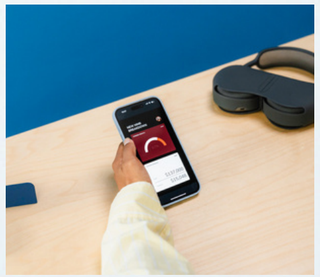
First Results

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Second Research

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Third Research

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RESULTS AND DISCUSSION

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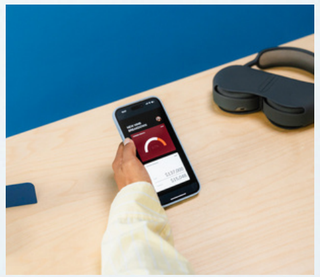
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CONCLUSION

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REFERENCES

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